Software Engineering

Topic 1: Introduction to Software Engineering (Questions 1-25)

1. What is software engineering?
A) Writing programs in high-level languages
B) The application of a systematic, disciplined, quantifiable approach to development, operation, and maintenance of software
C) The process of testing software
D) The design of computer hardware

Correct = B

2. Which of the following is NOT a type of software?
A) System software
B) Application software
C) Firmware software
D) Utility software

Correct = C

3. Generic software is designed for:
A) A specific customer only
B) A wide range of market with common requirements
C) Military applications only
D) Embedded systems only

Correct = B

4. What is a software crisis?
A) The rapid growth of software companies
B) Inability to develop software on time, within budget, and meeting requirements
C) The shortage of programmers
D) The high cost of hardware

Correct = B

5. Which of the following is a management myth?
A) Adding more programmers to a late project will catch it up
B) Requirements should be well-defined
C) Testing should be planned
D) Documentation is important

Correct = A

6. According to IEEE, software engineering is the application of:
A) Programming languages only
B) A systematic, disciplined, quantifiable approach to development, operation, and maintenance
C) Hardware design principles
D) Network protocols

Correct = B

7. What is a program?
A) A set of instructions written for a specific task + documentation
B) A set of instructions written for a specific task
C) A set of instructions + operating procedures
D) A set of instructions + user manuals

Correct = B

8. Software = ?
A) Program + documentation
B) Program + operating procedures
C) Program + documentation + operating procedures
D) Only source code

Correct = C

9. Which of the following is a limitation of software?
A) It never deteriorates but never wears out
B) It can be easily migrated
C) It is scalable
D) It is manufactured like hardware

Correct = A

10. Customized software is designed for:
A) Mass market
B) A specific customer with unique requirements
C) Operating systems only
D) Utility programs

Correct = B

11. Which type of software includes operating systems and compilers?
A) Application software
B) System software
C) Utility software
D) Business software

Correct = B

12. System software examples include:
A) Payroll system
B) Banking software
C) Operating system and compilers
D) MS-Office

Correct = C

13. Application software examples include:
A) Operating system
B) BIOS programs
C) Banking software and payroll system
D) Compilers

Correct = C

14. Utility software examples include:
A) Operating system
B) Banking software
C) MS-Office, compilers, debuggers
D) Embedded software

Correct = C

15. What is the role of software?
A) To design hardware
B) To perform tasks for users by activating and controlling hardware
C) To manufacture computers
D) To write documentation

Correct = B

16. Which of the following is an attribute of good software?
A) High cost
B) Maintainability
C) Complexity
D) Obscurity

Correct = B

17. The measure of good software is primarily:
A) Number of lines of code
B) Customer satisfaction
C) Use of latest technology
D) Number of programmers

Correct = B

18. Fritz Bauer (1969) defined software engineering as:
A) Writing programs in assembly language
B) Establishment and use of sound engineering principles for economically obtaining reliable software
C) Designing user interfaces
D) Managing software projects

Correct = B

19. Boehm's definition of software engineering emphasizes:
A) Hardware design
B) Scientific knowledge application to design and construction of programs
C) Network protocols
D) Database management

Correct = B

20. Software engineering activities include all EXCEPT:
A) Requirement analysis
B) Hardware manufacturing
C) Testing
D) Documentation

Correct = B

21. Which approach decomposes systems by functions in a structured manner?
A) OOSAD
B) SSAD
C) RAD
D) Agile

Correct = B

22. In OOSAD, objects have:
A) Only attributes
B) Only methods
C) Attributes, methods, and relationships
D) Only relationships

Correct = C

23. Software process fundamental activities include:
A) Hardware design
B) Specification, development, validation, evolution
C) Manufacturing
D) Distribution

Correct = B

24. Modern software development is complex because:
A) Hardware is expensive
B) It is technology driven and requires knowledge on different fronts
C) Programmers are scarce
D) Documentation is optional

Correct = B

25. Which of the following is NOT a software class?
A) Generic
B) Customized
C) Prototype
D) Both A and B

Correct = C

Topic 2: Software Development Life Cycle (SDLC) (Questions 26-45)

26. SDLC stands for:
A) Software Design Life Cycle
B) System Development Life Cycle
C) Software Development Life Cycle
D) Both B and C

Correct = B

27. How many phases typically constitute SDLC?
A) 3-4
B) 5-9
C) 10-12
D) Only 2

Correct = B

28. Which phase investigates information needs and determines resource requirements?
A) Design
B) Coding
C) Feasibility study
D) Testing

Correct = C

29. Technical feasibility confirms:
A) Profitability of the project
B) Hardware and software capability to satisfy needs
C) User willingness to operate
D) Safety requirements

Correct = B

30. Economic feasibility signifies:
A) Technical capability
B) Profitability with respect to cost saving and increased revenue
C) User acceptance
D) Safety of the system

Correct = B

31. Operational feasibility identifies:
A) Hardware requirements
B) Willingness and ability of users to operate and support
C) Safety constraints
D) Cost estimates

Correct = B

32. Behavioral feasibility is associated with:
A) Business applications
B) Real-time and embedded software development
C) Database design
D) User interface design

Correct = B

33. SRS stands for:
A) Software Requirement Specification
B) System Requirement Standard
C) Software Review System
D) System Review Specification

Correct = A

34. DDS stands for:
A) Data Design System
B) Design Document Specification
C) Development Document Standard
D) Detailed Design Solution

Correct = B

35. Which tools are used during design phase?
A) Compilers
B) Data Flow Diagram, Decision Table, Data Dictionary
C) Debuggers
D) Testing tools

Correct = B

36. Coding phase emphasizes:
A) Requirement gathering
B) Improvisation of programming efficiency
C) User acceptance
D) Feasibility analysis

Correct = B

37. Testing phase output includes:
A) SRS document
B) Test report and error report
C) User manual
D) Design document

Correct = B

38. Certification compares software performance against standards from:
A) Local companies
B) SEI, CMM, ISO
C) Hardware manufacturers
D) User groups

Correct = B

39. Implementation phase includes:
A) Requirement analysis
B) Site selection, file conversion, and final testing
C) Coding
D) Design

Correct = B

40. Which type of maintenance includes correction of errors?
A) Perfective
B) Adaptive
C) Corrective
D) Preventive

Correct = C

41. Review is conducted based on attributes like:
A) Code length
B) Ease of use, response time, reliability
C) Number of programmers
D) Hardware cost

Correct = B

42. Reverse engineering is applied when:
A) Building a new system from scratch
B) User needs modification of existing product
C) Feasibility report is accepted for new product
D) Testing is complete

Correct = B

43. Re-engineering enables:
A) Creating completely new systems
B) Better versions of existing product without changing core functionality
C) Writing new code
D) Hardware upgrades

Correct = B

44. The entry criteria of one phase becomes the _____ of the next phase.
A) Output
B) Exit criteria
C) Input
D) Both A and B

Correct = C

45. Which phase translates DDS into codes?
A) Design
B) Testing
C) Coding
D) Implementation

Correct = C

Topic 3: Software Process Models (Questions 46-85)

46. Waterfall model was proposed by:
A) Boehm
B) Winston Royce
C) Jacobson
D) Grady Booch

Correct = B

47. In which year was the Waterfall model proposed?
A) 1960
B) 1970
C) 1980
D) 1990

Correct = B

48. The Waterfall model is also known as:
A) Spiral model
B) Linear sequential model
C) Incremental model
D) Prototype model

Correct = B

49. Which phase in Waterfall model determines requirements for all system components?
A) Requirement analysis
B) System engineering
C) Design
D) Testing

Correct = A

50. SRS document describes:
A) How the system will be implemented
B) What the system will do without describing how
C) Code structure
D) Test cases

Correct = B

51. In traditional design approach, structured design consists of:
A) Only architectural design
B) Only detailed design
C) Architectural design and detailed design
D) Only coding

Correct = C

52. Alpha testing is conducted by:
A) Customers at their site
B) Development team at developer's site
C) Independent testers
D) End users

Correct = B

53. Beta testing is performed by:
A) Development team
B) Group of friendly customers
C) Regulators
D) Hardware engineers

Correct = B

54. Maintenance costs are often more than what percentage of total cost?
A) 20%
B) 40%
C) 60%
D) 80%

Correct = C

55. Perfective maintenance accounts for approximately what percentage of maintenance costs?
A) 24%
B) 19%
C) Over 50%
D) 10%

Correct = C

56. Prototyping model is based on:
A) Waterfall model
B) Iterative model
C) Spiral model
D) RAD model

Correct = B

57. Evolutionary prototyping involves:
A) Throwing away the prototype after use
B) Refining through repeated stages until adequate system developed
C) Only designing user interfaces
D) Skipping requirements

Correct = B

58. Throwaway prototyping is also called:
A) Evolutionary prototyping
B) Rapid prototyping
C) Incremental prototyping
D) Spiral prototyping

Correct = B

59. RAD model emphasizes:
A) Long development cycles
B) Short development cycles
C) Heavy documentation
D) Sequential phases only

Correct = B

60. RAD phases include all EXCEPT:
A) Business modeling
B) Data modeling
C) Process modeling
D) Hardware manufacturing

Correct = D

61. Incremental model consists of:
A) Single delivery
B) Stepwise development with expanding increments
C) No user feedback
D) Only one iteration

Correct = B

62. Spiral model was proposed by:
A) Royce
B) Boehm
C) Jacobson
D) Beck

Correct = B

63. Spiral model incorporates:
A) Only planning
B) Project risk factor
C) Only development
D) Only testing

Correct = B

64. The four activities in each phase of Spiral model are:
A) Design, code, test, deploy
B) Planning, risk analysis, development, assessment
C) Analysis, design, coding, testing
D) Elicitation, specification, validation, management

Correct = B

65. Component assembly model is based on:
A) Procedural programming
B) Object-oriented paradigm
C) Functional programming
D) Structured programming

Correct = B

66. Concurrent development model is also known as:
A) Sequential engineering
B) Concurrent engineering
C) Waterfall engineering
D) Prototype engineering

Correct = B

67. Formal methods model uses:
A) Natural language specifications
B) Mathematical specifications
C) Graphical models
D) Prototypes

Correct = B

68. 4GT stands for:
A) Fourth Generation Technology
B) Fourth Generation Techniques
C) Four Group Testing
D) Four Grade Tools

Correct = A

69. Which model is best suited for large projects with known, stable requirements?
A) Prototype model
B) Waterfall model
C) Spiral model
D) RAD model

Correct = B

70. Which model is preferable when requirements are not fully known?
A) Waterfall
B) Evolutionary

C) RAD
D) Formal methods

Correct = B

71. Which model is best for controlling project risks?
A) Waterfall
B) Prototype
C) Spiral
D) Incremental

Correct = C

72. RAD model requires:
A) Single team
B) Multiple teams working concurrently
C) No customer involvement
D) Long development time

Correct = B

73. In Spiral model, the radial dimension represents:
A) Progress
B) Cumulative cost
C) Time
D) Risk

Correct = B

74. In Spiral model, the angular dimension represents:
A) Cost
B) Progress of activity
C) Risk
D) Resources

Correct = B

75. Which model leads to software reuse?
A) Waterfall
B) Component assembly
C) Prototype
D) RAD

Correct = B

76. The Concurrent development model is particularly suited for:
A) Mainframe applications
B) Client/server applications
C) Embedded systems
D) Batch processing

Correct = B

77. Formal methods model enables:
A) Quick prototyping
B) Mathematical specification of software
C) User involvement
D) Rapid development

Correct = B

78. 4GT uses:
A) Assembly language
B) Non-procedural languages for database query
C) Machine language
D) Only C++

Correct = B

79. Selection of lifecycle model depends on:
A) Only requirements
B) Requirements, development team, users, project type and risks
C) Only team size
D) Only budget

Correct = B

80. When requirements change often, which model is suitable?
A) Waterfall
B) Prototype
C) RAD
D) Formal methods

Correct = B

81. When development team has less experience on similar projects, which model is suitable?
A) Waterfall
B) Prototype
C) RAD
D) Formal

Correct = B

82. When user involvement in all phases is required, which model is suitable?
A) Waterfall
B) Prototype
C) RAD
D) Spiral

Correct = B

83. When funding is stable, which model is suitable?
A) Waterfall
B) Prototype
C) Evolutionary
D) Spiral

Correct = A

84. When tight project schedule exists, which model is suitable?
A) Waterfall
B) Formal
C) Incremental, Evolutionary, Spiral, RAD
D) Only Waterfall

Correct = C

85. Phase containment of errors means:
A) Finding errors after delivery
B) Detecting errors as close to their point of introduction as possible
C) Ignoring errors
D) Testing only at the end

Correct = B

Topic 4: Software Project Management (Questions 86-135)

86. Software project management activities include:
A) Only coding
B) Planning, monitoring, and controlling
C) Only testing
D) Only documentation

Correct = B

87. SPMP stands for:
A) Software Process Management Plan
B) Software Project Management Plan
C) System Project Management Protocol
D) Software Program Management Plan

Correct = B

88. Which section in SPMP provides project overview?
A) Project organization
B) Introduction
C) Managerial process
D) Technical process

Correct = B

89. WBS stands for:
A) Work Breakdown Structure
B) Work Budget Schedule
C) Work Block System
D) Work Build Structure

Correct = B

90. WBS is a:
A) Testing technique
B) Hierarchical decomposition of project into successive levels
C) Programming language
D) Design pattern

Correct = B

91. PERT stands for:
A) Program Evaluation and Review Technique
B) Project Evaluation and Review Technique
C) Program Estimation and Review Technique
D) Project Estimation and Review Technique

Correct = B

92. CPM stands for:
A) Critical Process Method
B) Critical Path Method
C) Control Path Method
D) Critical Program Method

Correct = B

93. In network diagram, an event is represented by:
A) Arrow
B) Circle
C) Square
D) Triangle

Correct = B

94. An activity is represented by:
A) Circle
B) Arrow
C) Square
D) Diamond

Correct = B

95. Critical path is:
A) The shortest path
B) The sequence of activities deciding total project duration
C) Path with maximum float
D) Path with minimum activities

Correct = B

96. Float or slack is:
A) Time required for critical activities
B) Spare time a non-critical activity can consume without delaying project
C) Project duration
D) Activity cost

Correct = B

97. EST stands for:
A) Early Start Time
B) Earliest Start Time
C) Estimated Start Time
D) Event Start Time

Correct = B

98. LFT stands for:
A) Latest Finish Time
B) Last Finish Time
C) Late Finish Time
D) Limited Finish Time

Correct = B

99. In PERT, optimistic time is:
A) Longest possible time
B) Shortest possible time if everything goes exceptionally well
C) Most likely time
D) Average time

Correct = B

100. Pessimistic time is:
A) Shortest possible time
B) Time if mostly things go wrong
C) Normal expected time
D) Average time

Correct = B

101. In PERT, expected time formula is:
A) (t0 + tm + tp)/3
B) (t0 + 4tm + tp)/6
C) (t0 + tm + tp)/6
D) (t0 + 2tm + tp)/4

Correct = B

102. Gantt chart was developed by:
A) Boehm
B) Henry L. Gantt
C) Royce
D) Humphrey

Correct = B

103. Gantt chart shows:
A) Activities as bars on a time scale
B) Network relationships
C) Critical path
D) Risk factors

Correct = A

104. LOC stands for:
A) Lines of Code
B) Level of Complexity
C) Language of Code
D) Length of Code

Correct = A

105. Function Point metric was proposed by:
A) Boehm
B) Allan Albrecht
C) Halstead
D) McCabe

Correct = B

106. Function Point counts include all EXCEPT:
A) External inputs
B) External outputs
C) Internal processing time
D) External inquiries

Correct = C

107. FP = UFP × ?
A) 0.65
B) CAF (Complexity Adjustment Factor)
C) LOC
D) KLOC

Correct = B

108. CAF = ?
A) $0.65 + 0.01 \times \Sigma Fi$
B) $0.65 \times \Sigma Fi$
C) $0.01 + 0.65 \times \Sigma Fi$
D) $\Sigma Fi \times 0.65$

Correct = A

109. Feature Point metric extends Function Point by considering:
A) Lines of code
B) Algorithmic complexity
C) Number of programmers
D) Hardware requirements

Correct = B

110. COCOMO was proposed by:
A) Boehm
B) Halstead
C) McCabe
D) Putnam

Correct = A

111. COCOMO stands for:
A) Constructive Cost Model
B) Computer Cost Model
C) Constructive Code Model
D) Computer Code Model

Correct = A

112. Organic mode projects are typically:
A) System programs
B) Application programs with small team
C) Embedded systems
D) Real-time systems

Correct = B

113. Embedded mode projects are:
A) Application programs
B) System programs heavily coupled with hardware
C) Utility programs
D) Web applications

Correct = B

114. Basic COCOMO formula for effort is:
A) $E = a(KLOC)^b$
B) $E = a \times KLOC + b$
C) $E = a + b \times KLOC$
D) $E = KLOC^a \times b$

Correct = A

115. Intermediate COCOMO includes:
A) Only size
B) Size and cost drivers
C) Only cost drivers
D) Only time

Correct = B

116. EAF stands for:
A) Effort Adjustment Factor
B) Estimation Adjustment Factor
C) Effort Allocation Factor
D) Estimate Application Factor

Correct = A

117. COCOMO-II was developed at:
A) MIT
B) University of Southern California
C) Stanford
D) Carnegie Mellon

Correct = B

118. Halstead's software science metrics include:
A) Lines of code
B) Number of unique operators and operands
C) Function points
D) Feature points

Correct = B

119. Program volume in Halstead's metrics is:
A) $V = N \log_2 n$
B) $V = n \log_2 N$
C) $V = N \times n$
D) $V = N + n$

Correct = A

120. Cyclomatic complexity was developed by:
A) Halstead
B) McCabe
C) Boehm
D) Putnam

Correct = B

121. Cyclomatic complexity V(G) = ?
A) $E - N + 2$
B) $E + N - 2$
C) $E - N$
D) $E + N$

Correct = A

122. V(G) = P + 1, where P is:
A) Number of paths
B) Number of predicate nodes
C) Number of edges
D) Number of nodes

Correct = B

123. Rayleigh-Norden curve shows relationship between:
A) Cost and time
B) Effort applied and delivery time
C) Size and complexity
D) Quality and cost

Correct = B

124. Putnam's model formula is:
A) L = Ck × K^(1/3) × td^(4/3)
B) L = Ck × K × td
C) L = Ck × K^2 × td
D) L = Ck × K × td^2

Correct = A

125. In Putnam's model, Ck is:
A) Cost factor
B) Technological constant
C) Complexity factor
D) Time factor

Correct = B

126. Jensen's model is suitable for:
A) Very large systems
B) Small/medium size projects
C) Only embedded systems
D) Only real-time systems

Correct = A

127. Egoless teams are also called:
A) Hierarchical teams
B) Democratic teams
C) Chief programmer teams
D) Controlled teams

Correct = B

128. Chief programmer team consists of:
A) Only programmers
B) Chief programmer, backup programmer, librarian, and programmers
C) Only manager
D) Only testers

129. Controlled decentralized team combines strengths of:
A) Waterfall and agile
B) Democratic and chief programmer teams
C) Spiral and prototype
D) RAD and incremental

Correct = B

130. Software risk can be categorized as:
A) Only project risk
B) Project, product, and business risks
C) Only technical risk
D) Only schedule risk

Correct = B

131. Risk exposure = ?
A) Probability of risk × loss
B) Probability of risk + loss
C) Probability of risk - loss
D) Loss / probability

Correct = B

132. Risk control involves:
A) Only planning
B) Planning, mitigation, resolution, monitoring
C) Only monitoring
D) Only mitigation

Correct = B

133. Risk leverage = ?
A) (Risk exposure before - after) / Cost of reduction
B) (Risk exposure after - before) / Cost
C) Cost × risk exposure
D) Cost / risk exposure

Correct = A

134. In Rayleigh model, peak time td is related to 'a' by:
A) td^2 = 1/a
B) td^2 = 1/(2a)
C) td = a
D) td = 1/a

Correct = A

135. In Rayleigh model, cumulative manpower at start is:
A) Maximum
B) Zero
C) Constant
D) Infinity

Correct = B

Topic 5: Requirements Engineering (Questions 136-175)

136. According to IEEE Standard 729, a requirement is:
A) A condition or capability needed by a user to solve a problem
B) A program
C) A design document
D) A test case

Correct = A

137. Requirements are statements of:
A) How the system will be built
B) What the system will do
C) Which programming language to use
D) Hardware specifications

Correct = B

138. Problem analysis involves:
A) Writing code
B) Brainstorming and interviewing to understand the problem
C) Designing the system
D) Testing

Correct = B

139. Product description results in:
A) Code
B) SRS document
C) Test plan
D) User manual

Correct = B

140. Requirements elicitation is the process of:
A) Writing code
B) Getting requirements from stakeholders
C) Testing the system
D) Designing interface

Correct = Bs

141. Which is NOT a problem in requirements elicitation?
A) Scope problem
B) Understanding problem
C) Volatility problem
D) Coding problem

Correct = D

142. Requirements analysis involves checking for:
A) Only functionality
B) Consistency, incompleteness, ambiguity, priority
C) Only cost
D) Only schedule

Correct = B

143. Requirements specification can be:
A) Only written document
B) Written document, graphical model, formal model, use cases, or combination
C) Only code
D) Only prototypes

Correct = B

144. Requirements validation checks for:
A) Only consistency
B) Non-ambiguity, consistency, completeness, conformance to standards
C) Only completeness
D) Only testability

Correct = B

145. FAST stands for:
A) Fast Application Specification Technique
B) Facilitated Application Specification Techniques
C) Functional Application System Test
D) Formal Application Specification Tool

Correct = B

146. FAST involves:
A) Individual work
B) Joint team of customers and developers
C) Only customers
D) Only developers

Correct = B

147. Prototyping during requirements helps to:
A) Write final code
B) Determine feasibility, validate functions, uncover missing requirements
C) Test the system
D) Deploy the system

Correct = B

148. SRS serves as basis for:
A) Coding only
B) System testing and verification
C) Hardware design
D) Marketing

Correct = B

149. Which of the following should NOT be in SRS?
A) Functional requirements
B) Project requirements like staffing and costs
C) Non-functional requirements
D) Behavioral requirements

Correct = B

150. A high-quality SRS must be:
A) Ambiguous
B) Correct, unambiguous, verifiable, consistent, complete
C) Incomplete
D) Unverifiable

Correct = B

151. A requirement is verifiable if:
A) It is written in natural language
B) There exists a cost-effective process to check it
C) It is short
D) It uses technical terms

Correct = B

152. SRS is consistent if:
A) No subset of requirements conflict
B) All requirements are long
C) It uses many diagrams
D) It has many sections

Correct = A

153. Traceability in SRS means:
A) Requirements can be traced to code
B) Origin of each requirement is clear and can be referenced
C) Requirements are numbered
D) Requirements are short

Correct = B

154. Backward-from-requirements traceability implies:
A) Understanding which components satisfy each requirement
B) Knowing why every requirement exists
C) Every component references requirements
D) All documents reference SRS

Correct = B

155. Completeness in SRS requires:
A) Only functional requirements
B) Everything software is supposed to do is included, responses to all inputs, all pages numbered
C) Only non-functional requirements
D) Only design details

Correct = B

156. IEEE 830 standard provides:
A) Coding standards
B) Guidelines for writing SRS
C) Testing standards
D) Design standards

Correct = B

157. In IEEE 830, product perspective subsection should:
A) Describe functions
B) Put product into perspective with other products
C) List constraints
D) Describe users

Correct = B

158. Product functions subsection should:
A) Describe design
B) Provide summary of functions software will perform
C) List hardware requirements
D) Describe test cases

Correct = B

159. User characteristics subsection should describe:
A) Code structure
B) General characteristics of eventual users
C) Database design
D) Test data

Correct = B

160. General constraints subsection may include:
A) User names
B) Regulatory policies, hardware limitations, interfaces
C) Code examples
D) Test results

Correct = B

161. Functional requirements section should specify:
A) How inputs are transformed to outputs
B) Hardware specifications
C) Network protocols
D) Development schedule

Correct = A

162. External interface requirements include:
A) Only user interface
B) User, hardware, software, and communication interfaces
C) Only hardware interface
D) Only software interface

Correct = B

163. Performance requirements include:
A) Only static requirements
B) Static and dynamic numerical requirements
C) Only dynamic requirements
D) Only user interface requirements

Correct = B

164. Design constraints may include:
A) User preferences
B) Standards compliance and hardware limitations
C) Test cases
D) Code examples

Correct = B

165. Availability attribute specifies:
A) User interface design
B) Checkpoint, recovery, and restart requirements
C) Code structure
D) Programming language

Correct = B

166. Security attribute specifies:
A) User interface colors
B) Protection from accidental or malicious access
C) Database design
D) Network protocols

Correct = B

167. Maintainability attribute specifies:
A) User training
B) Requirements for software to be maintained
C) Hardware specifications
D) Test data

Correct = B

168. In SRS, the term TBD means:
A) To Be Deleted
B) To Be Determined
C) Test Before Delivery
D) Technical Business Document

Correct = B

169. SRS should exclude:
A) Functional requirements
B) Project requirements
C) Non-functional requirements
D) Interface requirements

Correct = B

170. Requirements management includes:
A) Only tracking changes
B) Identifying, controlling, and tracking requirements and changes
C) Only identifying requirements
D) Only writing requirements

Correct = B

171. Context-free questions help:
A) Write code
B) Understand problem, people, and solution
C) Design database
D) Test system

Correct = B

172. Viewpoints in requirements engineering:
A) Are unnecessary
B) Help structure requirements elicitation and analysis
C) Replace SRS
D) Are only for design

Correct = B

173. Domain requirements are derived from:
A) User preferences
B) Application domain rather than specific user needs
C) Hardware constraints
D) Programming language

Correct = B

174. Non-functional requirements often apply to:
A) Individual functions only
B) System as a whole
C) Only user interface
D) Only database

Correct = B

175. Requirements negotiation involves:
A) Writing code
B) Resolving conflicts and agreeing on compromise requirements
C) Testing
D) Designing

Correct = B

Topic 6: Software Design & Coding (Questions 176-225)

176. Software design transforms:
A) Code to SRS
B) SRS to DDS/SDD
C) Test cases to code
D) User manual to design

177. DDS/SDD is the major output of:
A) Requirements phase
B) Design phase
C) Testing phase
D) Maintenance phase

178. Interface design describes:
A) Internal algorithms
B) Interfaces between software system and external entities

C) Database structure
D) Code organization

179. Architectural design shows:
A) Detailed algorithms
B) Overall structure of system and component interactions
C) Test cases
D) User screens

Correct = B

180. Detailed design specifies:
A) Only interfaces
B) Internal elements, algorithms, and data structures of components
C) Only user interface
D) Only hardware requirements

Correct = B

181. Coupling measures:
A) Interaction within a module
B) Interconnection between different modules
C) Module size
D) Number of lines of code

Correct = B

182. Highly coupled modules have:
A) Weak interconnection
B) High dependency on each other
C) No interconnection
D) Low dependency

Correct = B

183. Data coupling is:
A) Highest coupling
B) Lowest coupling where data is passed through argument lists
C) Moderate coupling
D) No coupling

Correct = B

184. Stamp coupling occurs when:
A) Modules share global data
B) Modules communicate using composite data items
C) One module controls another
D) Modules share no data

Correct = B

185. Control coupling exists when:
A) Modules share data
B) Data from one module directs execution order in another
C) Modules are independent
D) Modules share global variables

Correct = B

186. Common coupling occurs when:
A) Modules share global data items
B) Modules pass data through parameters
C) One module branches into another
D) Modules are completely independent

Correct = B

187. Content coupling is:
A) Lowest coupling
B) Highest degree of coupling
C) Medium coupling
D) No coupling

Correct = B

188. Cohesion measures:
A) Interconnection between modules
B) Strength of relations within a module
C) Module size
D) Number of interfaces

Correct = B

189. Coincidental cohesion is:
A) Best type of cohesion
B) Worst type where tasks are loosely related
C) Medium cohesion
D) Functional cohesion

Correct = B

190. Logical cohesion occurs when:
A) Elements perform similar operations
B) Elements are executed in same time span
C) Elements are executed in sequence
D) Elements perform single function

Correct = A

191. Temporal cohesion exists when:
A) Elements perform similar operations
B) Elements are executed within certain time span
C) Elements are in sequence
D) Elements operate on same data

Correct = B

192. Procedural cohesion exists when:
A) Elements are executed in preset order
B) Elements are executed at same time
C) Elements are unrelated
D) Elements share data

Correct = A

193. Communicational cohesion exists when:
A) Elements are in sequence
B) Elements concentrate on single area of data structure
C) Elements are randomly collected
D) Elements perform similar operations

Correct = B

194. Sequential cohesion exists when:
A) Output of one element is input to next
B) Elements are executed at same time
C) Elements are unrelated
D) Elements perform similar operations

Correct = A

195. Functional cohesion is:
A) Worst type
B) Best type where elements cooperate to achieve single function
C) Medium type
D) Low type

Correct = B

196. Span of control refers to:
A) Number of lines in module
B) Number of subordinate modules controlled by calling module
C) Number of interfaces
D) Module complexity

Correct = B

197. Module size should ideally be:
A) As large as possible
B) Focused on single purpose with high cohesion
C) Over 1000 lines
D) Under 10 lines

Correct = B

198. IEEE recommended DDS content includes:
A) Only code
B) Introduction, design entities, attributes
C) Only test cases
D) Only user manua

Correct = B

199. User Interface Design golden rules by Mandel include:
A) Place user in control
B) Reduce user's memory load
C) Make interface consistent
D) All of the above

Correct = D

200. GUI stands for:
A) General User Interface
B) Graphical User Interface
C) Generic User Interface
D) Global User Interface

Correct = B

201. CUI stands for:
A) Character User Interface
B) Central User Interface
C) Command User Interface
D) Control User Interface

Correct = A

202. Command language-based interface requires user to:
A) Use mouse
B) Remember and type commands
C) Click icons
D) Use touch screen

Correct = B

203. Menu-based interface advantage is:
A) No need to remember syntax
B) Faster than command line
C) More secure
D) Uses less memory

Correct = A

204. Direct manipulation interface uses:
A) Commands
B) Icons or objects
C) Text menus
D) Voice commands

Correct = B

205. X-Window system is based on:
A) Peer-to-peer model
B) Client-server model
C) Mainframe model
D) Standalone model

Correct = B

206. In X-Window system, the server runs on:
A) Remote computer
B) User's computer
C) Both
D) Neither

Correct = B

207. Window manager in X-Window system is responsible for:
A) Low-level graphics
B) Window positioning, resizing, iconification
C) Network communication
D) File management

Correct = B

208. Function-oriented design focuses on:
A) Data structures
B) Functional aspects of software product
C) Objects
D) User interface

Correct = B

209. Jackson System Development (JSD) was created by:
A) Boehm
B) Michael A Jackson
C) Warnier
D) Orr

Correct = B

210. JSD distinguishes stages:
A) Only coding
B) Modeling, network, implementation
C) Only testing
D) Only design

Correct = B

211. Warnier-Orr design method is based on:
A) Objects
B) Data structures
C) Functions
D) Events

Correct = B

212. Object-oriented design focuses on:
A) Functions performed by program
B) Data to be manipulated by program
C) Hardware
D) Network

Correct = B

213. Reuse-based design accepts:
A) New development only
B) Existing partition of reusable modules
C) Only hardware reuse
D) Only code reuse

Correct = B

214. Coding standards are:
A) Optional
B) Conventions programmers follow to standardize code
C) Hardware specifications
D) Testing guidelines

Correct = B

215. Good programming style aids:
A) Only execution speed
B) Modifiability, transportability, readability, usability, efficiency
C) Only memory usage
D) Only security

Correct = B

216. Code inspections were popularized by:
A) Boehm
B) Fagan
C) McCabe
D) Halstead

Correct = B

217. Code inspection roles include:
A) Only author
B) Moderator, reader, recorder, author
C) Only manager
D) Only tester

Correct = B

218. Code review benefits include:
A) Finding bugs earlier
B) Knowledge sharing
C) Establishing quality metrics
D) All of the above

Correct = D

219. Verification proves that:
A) Product works according to specifications
B) User is satisfied
C) Cost is within budget
D) Schedule is met

Correct = A

220. Black box testing ignores:
A) Output
B) Internal structure of program
C) Input
D) Requirements

Correct = A

221. White box testing requires knowledge of:
A) Only output
B) Internal structure of software
C) Only user interface
D) Only documentation

Correct = B

222. Cohesion should be:
A) Minimized
B) Maximized
C) Zero
D) Ignored

Correct = B

223. Coupling should be:
A) Maximized
B) Minimized
C) Zero always
D) Ignored

Correct = B

224. Which cohesion is best?
A) Coincidental
B) Logical
C) Functional
D) Temporal

Correct = B

225. Which coupling is worst?
A) Data
B) Stamp
C) Content
D) Control

Correct = B

Topic 7: Software Testing (Questions 226-265)

226. Software testing is a process of:
A) Writing code
B) Evaluating system to verify it satisfies requirements or identify deviations
C) Designing interfaces
D) Gathering requirements

Correct = B

227. According to Myers (1979), testing is:
A) Writing test cases
B) Analyzing program with intent of finding errors
C) Running the program
D) Debugging

Correct = B

228. Testing principles include all EXCEPT:
A) Tests based on user needs
B) Tests should be planned carefully
C) Testing should be exhaustive
D) Pareto principle applies

Correct = C

229. Pareto principle in testing implies:
A) 80% errors traceable to 20% components
B) 50% errors traceable to 50% components
C) 20% errors traceable to 80% components
D) All errors equally distributed

Correct = A

230. Unit testing tests:
A) Entire system
B) Individual components
C) User acceptance
D) Performance

Correct = B

231. Integration testing tests:
A) Individual modules
B) Interfaces between modules
C) User interface
D) Documentation

232. System testing includes:
A) Only unit testing
B) Alpha, beta, and acceptance testing
C) Only integration testing
D) Only regression testing

Correct = B

233. Alpha testing is conducted at:
A) Customer site
B) Developer site by customers under developer guidance
C) Independent test lab
D) Remote location

Correct = B

234. Beta testing is conducted at:
A) Developer site
B) Customer site by selected customers
C) Both
D) Neither

Correct = B

235. Acceptance testing determines whether to:
A) Start coding
B) Accept or reject system delivery
C) Begin design
D) Write requirements

Correct = B

236. Black box testing focuses on:
A) Internal structure
B) Functional requirements
C) Code paths
D) Data structures

Correct = B

237. Black box testing is also called:
A) Structural testing
B) Functional testing
C) Glass box testing
D) Path testing

Correct = B

238. White box testing is also called:
A) Functional testing
B) Structural testing
C) Black box testing
D) Acceptance testing

Correct = B

239. Static testing strategies include:
A) Running the program
B) Systematic examination of program structure
C) Only performance testing
D) Only load testing

Correct = B

240. Debugging is:
A) Testing
B) Identifying, pointing out, and correcting bugs
C) Writing test cases
D) Designing

Correct = B

241. Brute force debugging involves:
A) Using formal methods
B) Appending print statements to display variable values
C) Backtracking
D) Using debugger

Correct = B

242. Backtracking debugging involves:
A) Running all tests
B) Backtracking from error symptom place
C) Using print statements
D) Using automated tools

Correct = B

243. Cause elimination debugging uses:
A) Random testing
B) Induction or deduction with binary partitioning
C) Print statements
D) Debugger only

Correct = B

244. Program slicing is similar to:
A) Black box testing
B) Backtracking
C) White box testing
D) Acceptance testing

Correct = B

245. Syntax error is:
A) Logical error
B) Violation of language rules
C) Runtime error
D) Design error

Correct = B

246. Logic error deals with:
A) Syntax violations
B) Incorrect data fields, out of range terms, invalid combinations
C) Compiler errors
D) Linker errors

Correct = B

247. Top-down integration uses:
A) Stubs
B) Drivers
C) Both stubs and drivers
D) Neither

Correct = A

248. Bottom-up integration uses:
A) Stubs
B) Drivers
C) Both stubs and drivers
D) Neither

Correct = B

249. Regression testing ensures:
A) New features work
B) Changes haven't introduced new errors
C) System is fast
D) Documentation is complete

Correct = B

250. Smoke testing is:
A) Unit testing
B) Rolling integration approach for shrink-wrapped software
C) Acceptance testing
D) Beta testing

Correct = B

251. Sandwich integration combines:
A) Top-down and bottom-up
B) Black box and white box
C) Unit and integration
D) Alpha and beta

Correct = B

252. Test case describes:
A) Only input
B) Input and expected output
C) Only expected output
D) Only preconditions

Correct = B

253. Test suite is:
A) Single test case
B) Set of test cases
C) Testing tool
D) Test environment

Correct = B

254. Verification answers:
A) Are we building the right product?
B) Are we building the product right?
C) Is the user satisfied?
D) Is the cost within budget?

Correct = B

255. Validation answers:
A) Are we building the product right?
B) Are we building the right product?
C) Is the code correct?
D) Is testing complete?

Correct = B

256. Fault is:
A) Human mistake
B) Representation of an error
C) Execution of error
D) Syntax error

Correct = B

257. Failure is:
A) Human mistake
B) Representation of error
C) Execution of fault
D) Syntax error

Correct = C

258. Error is:
A) Fault
B) Mistake
C) Discrepancy between actual and specified value
D) Failure

Correct = B

259. Component testing focuses on:
A) Interfaces between modules
B) Individual components in isolation
C) Entire system
D) User acceptance

Correct = B

260. Interface testing tests:
A) Individual modules
B) Coupling among modules
C) User interface
D) Documentation

Correct = C

261. Depth-first integration integrates components:
A) Horizontally
B) Vertically downward
C) Randomly
D) Only at same level

Correct = B

262. Breadth-first integration integrates components:
A) Vertically
B) Horizontally at each level
C) Randomly
D) Only at lowest level

Correct = B

263. Peak load test determines:
A) Storage capacity
B) System handling at peak processing demand
C) Recovery time
D) Documentation clarity

Correct = B

264. Recovery testing determines:
A) Response time
B) Ability to recover data after failure
C) Storage capacity
D) Performance

Correct = B

265. Human factors testing determines:
A) Code coverage
B) How users will use the system
C) Storage requirements
D) Network performance

Correct = B

Topic 8: Software Certification & Quality (Questions 266-295)

266. Software certification measures:
A) Only code length
B) Highlighted attributes of software product against benchmarks
C) Only cost
D) Only delivery time

Correct = B

267. Quality attributes include all EXCEPT:
A) Correctness
B) Reliability
C) User friendliness
D) Hardware cost

Correct = D

268. Verification confirms:
A) Output of one phase confirms to previous phase
B) Fully developed system confirms to requirements
C) User satisfaction
D) Budget compliance

Correct = A

269. Validation confirms:
A) Output of one phase confirms to previous
B) Fully developed system confirms to requirements
C) Code is error-free
D) Documentation is complete

Correct = B

270. SQA stands for:
A) Software Quality Assurance
B) System Quality Assessment
C) Software Quality Analysis
D) System Quality Assurance

Correct = A

271. SQA plan includes:
A) Only testing
B) Product introduction, product plans, process description, quality goals, risks
C) Only coding standards
D) Only user manual

Correct = B

272. McCall's quality factors include:
A) Only product operation
B) Product operation, product revision, product transition
C) Only product revision
D) Only product transition

Correct = B

273. Product operation quality factors include:
A) Maintainability
B) Correctness, reliability, efficiency, integrity, usability
C) Portability
D) Reusability

Correct = B

274. Product revision factors include:
A) Correctness
B) Maintainability, testability, flexibility
C) Portability
D) Inter-operability

Correct = B

275. Product transition factors include:
A) Correctness
B) Portability, reusability, inter-operability
C) Reliability
D) Usability

Correct = B

276. SEI-CMM was developed by:
A) IBM
B) Software Engineering Institute, Carnegie Mellon
C) Microsoft
D) Oracle

Correct = B

277. CMM Level 1 is:
A) Repeatable
B) Initial (chaotic)
C) Defined
D) Optimizing

Correct = B

278. CMM Level 2 is:
A) Initial
B) Repeatable
C) Defined
D) Managed

Correct = C

279. CMM Level 3 is:
A) Initial
B) Repeatable
C) Defined
D) Optimizing

Correct = C

280. CMM Level 4 is:
A) Initial
B) Repeatable
C) Managed
D) Optimizing

Correct = C

281. CMM Level 5 is:
A) Initial
B) Repeatable
C) Defined
D) Optimizing

Correct = D

282. ISO stands for:
A) International Standards Organization
B) International Organization for Standardization
C) Integrated Systems Organization
D) Internet Standards Organization

Correct = B

283. ISO 9001 specifies:
A) Coding standards
B) Quality assurance elements applicable to any business
C) Testing standards
D) Design standards

Correct = B

284. ISO 9126 identifies software quality characteristics including:
A) Only functionality
B) Functionality, reliability, usability, efficiency, maintainability, portability
C) Only reliability
D) Only portability

Correct = B

285. Reliability is defined as:
A) Probability of failure-free operation in specified environment for specified time
B) System speed
C) User satisfaction
D) Code size

Correct = A

286. MTTF stands for:
A) Mean Time To Failure
B) Mean Time To Fix
C) Maximum Time To Failure
D) Minimum Time To Failure

Correct = A

287. MTTR stands for:
A) Mean Time To Repair
B) Mean Time To Restart
C) Maximum Time To Repair
D) Minimum Time To Restart

Correct = B

288. MTBF = ?
A) MTTF + MTTR
B) MTTF - MTTR
C) MTTF × MTTR
D) MTTF / MTTR

Correct = A

289. Availability = ?
A) MTTF / (MTTF + MTTR) × 100%
B) MTTR / (MTTF + MTTR) × 100%
C) MTTF × MTTR
D) MTTF - MTTR

Correct = A

290. Jelinski-Moranda model is a:
A) Design model
B) Reliability growth model
C) Testing model
D) Cost model

Correct = B

291. PSP stands for:
A) Personal Software Process
B) Program Software Process
C) Project Software Process
D) Process Software Program

Correct = B

292. Six Sigma aims to eliminate defects per million opportunities to:
A) 1000
B) 3-4
C) 10000
D) 0

Correct = B

293. DMAIC stands for:
A) Define, Measure, Analyze, Improve, Control
B) Design, Model, Apply, Implement, Control
C) Define, Model, Analyze, Implement, Control
D) Design, Measure, Analyze, Improve, Control

Correct = A

294. DMADV stands for:
A) Define, Measure, Analyze, Design, Verify
B) Design, Model, Apply, Develop, Verify
C) Define, Model, Analyze, Develop, Verify
D) Design, Measure, Analyze, Develop, Verify

Correct = A

295. Personal Software Process emphasizes:
A) Team management
B) Time measurement and planning
C) Hardware design
D) Network configuration

Correct = B

Topic 9: Software Maintenance & Evolution (Questions 296-325)

296. Software maintenance describes activities that:
A) Precede delivery
B) Follow delivery of initial complete version
C) Only involve bug fixing
D) Only involve documentation

Correct = B

297. Approximately what percentage of software project budget is spent on maintenance?
A) 20%
B) 40%
C) 70%
D) 10%

Correct = C

298. Corrective maintenance refers to:
A) Enhancing performance
B) Repairing processing or performance failures
C) Adapting to environmental change
D) Preventing obsolescence

Correct = B

299. Adaptive maintenance means:
A) Fixing bugs
B) Changing program function to adapt to environmental change
C) Improving performance
D) Preventing obsolescence

Correct = B

300. Perfective maintenance means:
A) Fixing bugs
B) Enhancing performance or modifying programs for user needs
C) Adapting to new hardware
D) Preventing obsolescence

Correct = B

301. Preventive maintenance involves:
A) Bug fixing
B) Reengineering and reverse engineering to prevent obsolescence
C) Performance enhancement
D) User training

Correct = B

302. Maintenance challenges include all EXCEPT:
A) Understanding program written by others
B) High staff turnover
C) Systems designed for change
D) Inadequate documentation

Correct = C

303. Build and Fix model is:
A) Structured approach
B) Ad hoc approach to maintain software
C) Formal method
D) Iterative approach

Correct = B

304. Iterative enhancement model has stages:
A) Only coding
B) Analysis, characterization of modification, redesign and implementation
C) Only testing
D) Only design

Correct = B

305. Reuse-oriented model steps include:
A) Only coding
B) Identification, analysis, modification, integration of reusable parts
C) Only testing
D) Only design

Correct = B

306. Boehm's maintenance model is based on:
A) Waterfall principles
B) Economic model principles
C) Agile principles
D) Prototype principles

Correct = B

307. Taute maintenance model has how many phases?
A) 4
B) 6
C) 8
D) 10

Correct = C

308. In Taute model, Change request phase:
A) Estimates time
B) Receives change request from client
C) Modifies source code
D) Tests the code

Correct = B

309. In Taute model, Estimate phase:
A) Receives request
B) Estimates time and effort for modification
C) Modifies code
D) Tests code

Correct = B

310. In Taute model, Programming phase:
A) Estimates effort
B) Modifies source code
C) Tests code
D) Delivers product

Correct = B

311. Belady and Lehman model for maintenance cost is:
A) M = P + Keˆ(c−d)
B) M = P × Keˆ(c−d)
C) M = P − Keˆ(c−d)
D) M = P / Keˆ(c−d)

Correct = A

312. Lehman's first law states:
A) Software structure degrades over time
B) Software must change continually or become less useful
C) Rate of development is constant
D) Software never changes

Correct = B

313. Lehman's second law states:
A) Software must change continually
B) Structure of program tends to degrade as maintenance is carried out
C) Rate of development is constant
D) Software never degrades

Correct = B

314. Lehman's third law states:
A) Software must change
B) Structure degrades
C) Rate of development is approximately constant
D) Software is static

Correct = C

315. ACT stands for:
A) Annual Change Traffic
B) Actual Change Time
C) Activity Change Test
D) Application Change Tool

Correct = A

316. Annual Maintenance Effort (AME) = ?
A) ACT × SDE
B) ACT + SDE
C) ACT / SDE
D) SDE / ACT

317. Software configuration management deals with:
A) Only code
B) Tracking and controlling configuration of software product during life cycle
C) Only documentation
D) Only testing

Correct = B

318. Configuration identification includes:
A) Only controlled objects
B) Controlled, pre-controlled, and uncontrolled objects
C) Only uncontrolled objects
D) Only deliverables

Correct = B

319. Baseline in configuration management is:
A) Initial version
B) Frozen updated version under configuration control
C) Final version
D) Test version

Correct = B

320. Configuration control board (CCB) evaluates:
A) Code quality
B) Change requests
C) Test results
D) User satisfaction

Correct = B

321. Status accounting records:
A) Only what happened
B) What happened, who did it, when it happened, what will be affected
C) Only who did it
D) Only when it happened

Correct = B

322. Configuration audit ensures:
A) Code is written
B) Changes made as per specification and properly documented
C) User is satisfied
D) Budget is met

Correct = B

323. Version control tool keeps record of:
A) Only code
B) Name, version, compiler used, staff, date, time
C) Only date
D) Only staff name

Correct = B

324. Evolutionary graph in version control shows:
A) Code execution
B) Evolution of configuration item during development life cycle
C) Test results
D) User feedback

Correct = B

325. Commercial version control tools include:
A) Only Microsoft Word
B) Rational Clear Case, Microsoft Visual Source Safe, Git
C) Only Excel
D) Only PowerPoint

Correct = B

Topic 10: Software Re-engineering (Questions 326-340)

326. Re-engineering is:
A) Building new system from scratch
B) Modifying internal mechanism without changing functionality
C) Fixing bugs only
D) Writing new documentation

Correct = B

327. Re-engineering process model includes:
A) Only coding
B) Inventory analysis, document restructuring, reverse engineering, code restructuring, data restructuring, forward engineering
C) Only testing
D) Only design

Correct = B

328. Inventory analysis involves:
A) Writing code
B) Creating inventory of all software products with description
C) Testing
D) Deploying

Correct = B

329. Document restructuring options include:
A) Only full documentation
B) Leave as is, update partially, or fully redocument
C) Only leaving as is
D) Only deleting

Correct = B

330. Reverse engineering is:
A) Writing new code
B) Analyzing program to create representation at higher level of abstraction
C) Testing
D) Deploying

Correct = B

331. Reverse engineering process includes:
A) Only coding
B) Restructure code, abstraction, producing specification
C) Only testing
D) Only design

Correct = B

332. Forward engineering is also called:
A) Reverse engineering
B) Renovation or reclamation
C) Maintenance
D) Testing

Correct = B

333. Redocumentation produces:
A) New code
B) Exact documentation at same level of abstraction
C) New design
D) New tests

Correct = B

334. Design recovery identifies:
A) Only code errors
B) Meaningful higher-level abstractions from source code
C) Only syntax errors
D) Only runtime errors

Correct = B

335. Reverse engineering tools include:
A) Only compiler
B) Restructurer, cross-referencer, static analyzer, text editor
C) Only debugger
D) Only linker

Correct = B

336. Code restructuring improves:
A) Functionality
B) Code understandability, testability, and maintainability
C) Hardware performance
D) Network speed

Correct = B

337. Data restructuring involves:
A) Only code changes
B) Redefining data models, objects, and structures
C) Only hardware changes
D) Only user interface changes

Correct = B

338. Advantages of re-engineering include:
A) Higher cost
B) Lower cost, lower risk, efficient use of existing staff, incremental development
C) Longer time
D) More complex

Correct = B

339. Reverse engineering vs. forward engineering:
A) Both are same
B) Reverse derives design from code; forward generates system from design
C) Reverse is newer
D) Forward is older

Correct = B

340. Levels of reverse engineering include:
A) Only code level
B) Redocumentation, structural redocumentation, design recovery
C) Only design level
D) Only requirements level

Correct = B

Topic 11: Computer Aided Software Engineering (CASE) (Questions 341-360)

341. CASE stands for:
A) Computer Aided Software Engineering
B) Computer Assisted Software Environment
C) Computer Aided System Engineering
D) Computer Application Software Engineering

Correct = A

342. CASE tools aid in:
A) Only coding
B) Translation of requirements to design, design to code, testing, documentation
C) Only testing
D) Only documentation

Correct = B

343. CASE architecture includes:
A) Only repository
B) User interface, tools management, object management system, repository
C) Only tools
D) Only database

Correct = B

344. Object Management System (OMS) maps:
A) Only data
B) Logical entities into underlying storage management
C) Only code
D) Only user interface

Correct = B

345. CASE repository is a:
A) Code storage
B) System developer database
C) User manual
D) Test suite

Correct = B

346. Upper CASE tools support:
A) Implementation and maintenance
B) Analysis and design phases
C) Only coding
D) Only testing

Correct = B

347. Lower CASE tools support:
A) Analysis and design
B) Implementation and maintenance phases
C) Only requirements
D) Only planning

Correct = B

348. Integrated CASE tools include:
A) Only front-end features
B) Front-end and back-end development features
C) Only back-end features
D) Only testing tools

Correct = B

349. Reverse engineering tools help in:
A) Forward engineering
B) Analyzing existing application for higher level representation
C) Code generation
D) Testing

Correct = B

350. CASE advantages include:
A) Higher cost
B) Improvisation of productivity, simplified documentation, higher quality management
C) More errors
D) Longer time

Correct = B

351. CASE disadvantages include:
A) Low cost
B) High procurement cost, learning curve, mixture of tools
C) Easy to use
D) Quick to learn

Correct = B

352. Production Process Support Technology includes:
A) Only management
B) Support for specification, design, implementation, testing
C) Only planning
D) Only documentation

Correct = B

353. Meta-CASE technology includes:
A) Testing tools
B) Generators to create production process and management support tools
C) Compilers
D) Debuggers

Correct = B

354. Interface generator provides:
A) Code
B) Capability to prepare mock-ups and prototypes of user interfaces
C) Database
D) Network

Correct = B

355. Tools Management Services control:
A) Hardware
B) Behavior of tools within environment
C) User access
D) Network

Correct = B

356. CASE supports prototyping by:
A) Generating code only
B) Defining user interaction, system control flow, storing/retrieving data
C) Testing only
D) Documentation only

Correct = B

357. CASE supports structured analysis by:
A) Only drawing diagrams
B) Making complex diagrams and checking incompleteness across levels
C) Only storing data
D) Only generating reports

Correct = B

358. CASE supports code generation by:
A) Only compiling
B) Generating module skeletons, records, structures, class definitions
C) Only linking
D) Only debugging

Correct = B

359. CASE test case generator features include:
A) Only running tests
B) Supporting design and requirement testing, generating test set reports
C) Only debugging
D) Only compiling

Correct = B

360. CASE improves:
A) Only speed
B) Productivity, information system quality, effectiveness
C) Only documentation
D) Only testing

Correct = B

Topic 12: Unified Modeling Language (UML) (Questions 361-400)

361. UML stands for:
A) Unified Modeling Language
B) Universal Modeling Language
C) Unified Markup Language
D) Universal Markup Language

Correct = B

362. UML was released by OMG in:
A) 1987
B) 1997
C) 2000
D) 2005

Correct = B

363. Primary authors of UML include:
A) Boehm, McCabe, Halstead
B) Rumbaugh, Jacobson, Booch
C) Royce, Brooks, Humphrey
D) Fagan, Myers, Pressman

Correct = B

364. UML is a:
A) Programming language
B) Graphical language for visualizing, specifying, constructing, documenting
C) Database language
D) Testing language

Correct = B

365. UML architecture views include:
A) Only use case view
B) Use case, design, process, implementation, deployment views
C) Only design view
D) Only deployment view

Correct = B

366. Use case view describes:
A) System structure
B) Behavior of system by users, analysts, and testers
C) Hardware configuration
D) Database design

Correct = B

367. Design view describes:
A) Only user interface
B) Classes, interfaces, collaborations forming vocabulary
C) Only database
D) Only network

Correct = B

368. Process view describes:
A) Classes and interfaces
B) Threads and processes for concurrency and synchronization
C) Only user interface
D) Only hardware

Correct = B

369. Implementation view describes:
A) Only classes
B) Components and files used to assemble
C) Only processes
D) Only nodes

Correct = B

370. Deployment view encompasses:
A) Classes and objects
B) Nodes forming hardware topology
C) Only processes
D) Only use cases

Correct = B

371. Structural things in UML include:
A) Interactions
B) Class, interface, collaboration, use case, active class, component, node
C) State machines
D) Packages

Correct = B

372. Behavioral things in UML include:
A) Class and interface
B) Interaction and state machine
C) Component and node
D) Package

Correct = B

373. Grouping things in UML is:
A) Class
B) Package
C) Interface
D) Node

Correct = B

374. Relationships in UML include:
A) Only association
B) Dependency, association, generalization, realization
C) Only inheritance
D) Only composition

Correct = B

375. Use case diagram shows:
A) Class relationships
B) Functionality provided by system in terms of actors and use cases
C) State changes
D) Object interactions

Correct = B

376. Actor in use case diagram is represented by:
A) Rectangle
B) Stick figure
C) Ellipse
D) Arrow

Correct = B

377. Use case in diagram is represented by:
A) Stick figure
B) Ellipse
C) Rectangle
D) Diamond

Correct = B

378. Include relationship between use cases means:
A) One use case may extend another
B) Behavior of included use case is inserted into including use case
C) Use cases are independent
D) One use case generalizes another

Correct = B

379. Extend relationship between use cases means:
A) Behavior may be inserted under some conditions
B) One includes another
C) They are the same
D) They are unrelated

Correct = A

380. Class diagram shows:
A) Object interactions
B) Classes and relationships
C) State changes
D) Use cases

Correct = B

381. Class visibility symbols:
A) +, -, # represent public, private, protected
B) *, /, % represent types
C) @, &, $ represent methods
D) !, ?, ˆ represent attributes

Correct = A

382. Association represents:
A) Inheritance
B) Structural relationships between classes
C) Dependency
D) Realization

Correct = B

383. Cardinality in association:
A) 1 means zero or one
B) * means from zero to any positive integer
C) 0..1 means one and only one
D) 1..* means zero or one

Correct = B

384. Qualified association uses:
A) A diamond
B) A qualifier rectangle to partition set of objects
C) A dashed line
D) A triangle

385. Aggregation indicates:
A) Inheritance
B) Whole-part relationship
C) Dependency
D) Realization

Correct = B

386. Composition means:
A) Weak relationship
B) Part is member of only one composite object with existence dependency
C) No relationship
D) Temporary relationship

Correct = B

387. Realization connects:
A) Two classes
B) Class to interface that supplies behavioral specification
C) Two interfaces
D) Class to itself

Correct = B

388. Interface in UML is used to:
A) Define data
B) Describe visible behavior of class/component/package
C) Define attributes
D) Define methods only

Correct = B

389. Template class is:
A) Concrete class
B) Generic class parameterized for instantiation
C) Abstract class
D) Interface

Correct = B

390. Abstract class:
A) Can be instantiated
B) Cannot be instantiated directly
C) Has only methods
D) Has only attributes

Correct = B

391. Stereotype in UML is denoted by:
A) << >>
B) [[ ]]
C) {{ }}
D) (( ))

Correct = B

392. Sequence diagram shows:
A) Class relationships
B) Interactions among objects in time sequence
C) State changes
D) Component organization

Correct = B

393. Collaboration diagram shows:
A) Time sequence
B) Flow of control by organization
C) State changes
D) Use cases

Correct = B

394. Statechart diagram models:
A) Class structure
B) Event-ordered behavior of single use case or reactive objects
C) Component organization
D) Deployment

Correct = B

395. Activity diagram models:
A) Class structure
B) Workflow of single use case or entire software
C) Component organization
D) Node configuration

Correct = B

396. Component diagram shows:
A) Class relationships
B) Static implementation view with components and dependencies
C) State changes
D) Object interactions

Correct = B

397. Deployment diagram shows:
A) Class structure
B) Physical topology of processors, devices, and software components
C) Use cases
D) Object interactions

Correct = B

398. Package in UML is:
A) Class
B) Grouping mechanism for organizing elements
C) Interface
D) Component

Correct = B

399. OCL stands for:
A) Object Constraint Language
B) Object Class Language
C) Object Communication Language
D) Object Control Language

Correct = A

400. OCL is used to:
A) Write code
B) Express side-effect-free constraints within UML models
C) Design database
D) Test system

Correct = B

Topic 13: Object Oriented Software Engineering (Questions 401-435)

401. Object-oriented approach treats data as:
A) Secondary element
B) Critical element in program development
C) Optional
D) Only for output

Correct = B

402. Object is:
A) Function
B) Person, place, thing, concept, event bundling data and methods
C) Variable
D) Constant

Correct = B

403. Class defines:
A) Individual object
B) General structure of similar objects
C) Only methods
D) Only data

Correct = B

404. Attribute is:
A) Function
B) Data item defined as part of class or object
C) Method
D) Message

Correct = B

405. Method is:
A) Data item
B) Procedure or function defined as part of class
C) Object
D) Attribute

Correct = B

406. Message is:
A) Data item
B) Instruction to invoke action on object
C) Class
D) Attribute

Correct = B

407. Encapsulation is:
A) Hiding data only
B) Packaging data and operations into object
C) Inheritance
D) Polymorphism

Correct = B

408. Inheritance is:
A) Packaging data and operations
B) Defining new class in terms of existing class
C) Hiding data
D) Multiple methods same name

Correct = B

409. Polymorphism is:
A) Multiple inheritance
B) Method applied to objects of different types
C) Data hiding
D) Class definition

Correct = B

410. Overloading is:
A) Runtime polymorphism
B) Compile-time polymorphism with same name different parameters
C) Data hiding
D) Inheritance

Correct = B

411. Overriding is:
A) Compile-time polymorphism
B) Runtime polymorphism where subclass overrides superclass method
C) Data hiding
D) Multiple inheritance

Correct = B

412. OOA stands for:
A) Object-Oriented Analysis
B) Object-Oriented Architecture
C) Object-Oriented Application
D) Object-Oriented Algorithm

Correct = A

413. OOD stands for:
A) Object-Oriented Design
B) Object-Oriented Development
C) Object-Oriented Data
D) Object-Oriented Debugging

Correct = B

414. OOP stands for:
A) Object-Oriented Process
B) Object-Oriented Programming
C) Object-Oriented Protocol
D) Object-Oriented Package

Correct = B

415. OORA stands for:
A) Object-Oriented Requirements Analysis
B) Object-Oriented Real-time Architecture
C) Object-Oriented Runtime Analysis
D) Object-Oriented Rapid Application

Correct = A

416. Object-oriented SDLC phases include:
A) Only coding
B) OORA, OOA, OOD, OOP
C) Only design
D) Only testing

Correct = B

417. In OOA, sources for analysis include:
A) Only code
B) Written requirements, vision document, interviews
C) Only test cases
D) Only user manual

Correct = B

418. OOA result is:
A) Executable code
B) Conceptual model with use cases, class diagrams, interaction diagrams
C) Test plan
D) User manual

Correct = B

419. OOD transforms:
A) Code to design
B) Conceptual model to implementation classes and interfaces
C) Requirements to code
D) Tests to code

Correct = B

420. Benefits of object-oriented software include:
A) Higher cost
B) Managed complexity, improved quality, increased reusability, increased extensibility
C) More errors
D) Longer time

Correct = B

421. Demerits of object-oriented software include:
A) Easy to learn
B) Requires greater concentration on requirements, up-front investment, more testing
C) Quick to develop
D) No training needed

Correct = B

422. OOA vs OOD:
A) Both are same
B) OOA identifies requirements; OOD provides implementation
C) OOA is after OOD
D) OOD identifies requirements

Correct = B

423. OOPS programming languages include:
A) Only C
B) C++, Smalltalk, Java
C) Only FORTRAN
D) Only COBOL

Correct = B

424. Smalltalk is:
A) Procedural language
B) Pure object-oriented language
C) Functional language
D) Assembly language

Correct = B

425. Java is a mixture of:
A) C and FORTRAN
B) C++ and Smalltalk
C) COBOL and Pascal
D) LISP and Prolog

Correct = B

426. Object has:
A) Only state
B) State, behavior, and identity
C) Only behavior
D) Only identity

Correct = B

427. Data abstraction allows:
A) Hiding all data
B) Essential operations and data public without background details
C) Only methods public
D) Only attributes public

Correct = B

428. Data hiding makes encapsulated data:
A) Public
B) Inaccessible and invisible to other objects
C) Accessible to all
D) Global

Correct = B

429. Strong typing means:
A) Data types determined at runtime
B) Exact data type of arguments defined in program
C) No types
D) Weak types

Correct = B

430. Object persistence is:
A) Object deletion
B) Property extending object lifetime
C) Object creation
D) Object copying

Correct = B

431. Meta class is:
A) Class of objects
B) Class of classes
C) Subclass
D) Superclass

Correct = B

432. In object-oriented systems, objects communicate through:
A) Shared memory
B) Messages
C) Global variables
D) Direct access

Correct = B

433. Sequential operation in objects means:
A) All objects run in parallel
B) Message causes other messages to complete before return
C) No messages
D) Random execution

Correct = B

434. Instance variables can only be accessed by:
A) Other objects directly
B) Instance methods of same object
C) Global functions
D) Any method

Correct = B

435. Reusability in OOP is enhanced by:
A) Procedural programming
B) Inheritance and polymorphism
C) Global variables
D) Monolithic code

Correct = B

Topic 14: Software & Tools (Questions 436-465)

436. Analysis tools include:
A) Compilers
B) Data collection tools, charting tools, dictionary tools
C) Linkers
D) Debuggers

Correct = B

437. Data collection tools capture:
A) Code
B) Details describing current systems and procedures
C) Test results
D) User feedback

Correct = B

438. Charting tools create:
A) Code
B) Graphic representations of systems and activities
C) Executable files
D) Databases

Correct = B

439. Dictionary tools record and maintain:
A) Code
B) Descriptions of system elements
C) Test data
D) User names

Correct = B

440. Design tools include:
A) Compilers
B) Specification tools and layout tools
C) Debuggers
D) Linkers

Correct = B

441. Specification tools assist in:

A) Coding
B) Documenting features to be included in application
C) Testing
D) Deployment

Correct = B

442. Layout tools describe:
A) Code structure
B) Position of data, messages, headings on screens and reports
C) Algorithms
D) Network configuration

Correct = B

443. Development tools include:
A) Only compilers
B) Software engineering tools and code generators
C) Only debuggers
D) Only text editors

Correct = B

444. Code generators produce:
A) Documentation
B) Source code from functional specifications
C) Test cases
D) User manuals

Correct = B

445. Decision tree presents:
A) Code structure
B) Conditions and actions sequentially
C) Data flow
D) Class hierarchy

Correct = B

446. Decision table is segmented into quadrants:
A) Only condition statement
B) Condition statement, condition entry, action statement, action entry
C) Only action statement
D) Only condition entry

Correct = B

447. In limited entry decision table, Y denotes:
A) Maybe
B) Yes
C) No
D) Don't care

Correct = B

448. In limited entry decision table, – denotes:
A) Yes
B) No
C) Don't care
D) Perform action

Correct = C

449. Decision table advantages include:
A) Difficult to use
B) Clear structure, consistent decision making, easier documentation
C) No documentation
D) Complex to modify

Correct = B

450. DFD stands for:
A) Data Flow Diagram
B) Design Flow Diagram
C) Data Function Diagram
D) Design Function Diagram

Correct = A

451. DFD is a:
A) Testing tool
B) Graphic tool to describe and analyze movement of data through system
C) Programming tool
D) Debugging tool

Correct = B

452. In DFD, data flow is represented by:
A) Circle
B) Arrow
C) Rectangle
D) Diamond

Correct = B

453. In DFD, process is represented by:
A) Arrow
B) Circle or rounded rectangle
C) Open rectangle
D) Diamond

Correct = B

454. In DFD, source or destination is represented by:
A) Circle
B) Rectangle
C) Arrow
D) Diamond

Correct = B

455. In DFD, data store is represented by:
A) Circle
B) Open rectangle (two parallel lines)
C) Arrow
D) Diamond

Correct = B

456. Level 0 DFD is called:
A) Detailed DFD
B) Context level DFD
C) Process DFD
D) Data DFD

Correct = B

457. Data dictionary is a:
A) Code repository
B) Catalog of elements in system
C) Test suite
D) User manual

Correct = B

458. In data dictionary, = means:
A) Add
B) Is equivalent to
C) Optional
D) Iteration

Correct = B

459. In data dictionary, + means:
A) Is equivalent to
B) And
C) Or
D) Optional

Correct = B

460. In data dictionary, [ ] means:
A) And
B) Either or
C) Iteration
D) Optional

Correct = B

461. In data dictionary, { } means:
A) And
B) Either or
C) Iteration
D) Optional

Correct = C

462. Structured analysis steps include:
A) Only coding
B) Create DFD, develop data dictionary, write PSPECs
C) Only testing
D) Only deployment

Correct = B

463. Structure chart shows:
A) Data flow
B) Hierarchy of and interrelationships among modules
C) User interface
D) Database design

Correct = B

464. Transform analysis identifies:
A) Only output
B) Central transform after eliminating input and output streams
C) Only input
D) Only errors

Correct = B

465. Transaction analysis is based on:
A) Data flow
B) Different types of transactions system must process
C) Object hierarchy
D) User interface

Correct = B

Topic 14: Software & Tools (Questions 466-485)

466. In DFD, leveling means:
A) Adding more data stores
B) Representing system at different levels of abstraction
C) Removing processes
D) Adding more external entities

Correct = B

467. Physical DFDs show:
A) Logical data flow
B) Implementation details of the system
C) Only processes
D) Only data stores

Correct = B

468. Logical DFDs focus on:
A) Physical devices
B) Business functions and logical data flow
C) File formats
D) Hardware specifications

Correct = B

469. PSPEC (Process Specification) describes:
A) Data structures
B) Processing logic of primitive processes
C) User interface
D) Network configuration

470. Structured English for PSPEC uses:
A) Complex sentences
B) Sequence, if-then, do-while constructs only
C) Adverbs and adjectives
D) Technical jargon

Correct = B

471. Decision tables are preferred over flowcharts when:
A) Simple conditions exist
B) Complex decision logic with multiple conditions needs recording
C) Only sequential flow is needed
D) User interaction is high

Correct = B

472. Karnaugh map is used to check decision table for:
A) Code errors
B) Completeness and multiply specified actions
C) Performance issues
D) User satisfaction

Correct = B

473. Structured chart symbols include all EXCEPT:
A) Module with incoming/outgoing data arrows
B) Conditional execution diamond
C) Iteration arc
D) Data store symbol

Correct = D

474. In structure chart, arrow with transparent circle represents:
A) Control flag
B) Data element
C) Module call
D) Library routine

Correct = B

475. In structure chart, arrow with opaque circle represents:
A) Data element
B) Control flag
C) Module
D) Library

Correct = B

476. Data structure diagram shows:
A) Code structure
B) Logical data structure requirements
C) Network topology
D) User interface

Correct = B

477. In data structure diagram, entities are represented by:
A) Circles
B) Rectangles
C) Arrows
D) Diamonds

Correct = B

478. Attribute pointer in data structure diagram links:
A) Two entities by common information
B) Entities to processes
C) Data to code
D) Users to systems

Correct = B

479. Logical pointer in data structure diagram identifies:
A) Physical address
B) Relationships between entities
C) Memory location
D) File path

Correct = B

480. Fourth Generation Languages (4GL) emphasize:
A) How to perform a task
B) What is to be done rather than how
C) Machine code
D) Assembly language

Correct = B

481. Application generators permit:
A) Only report generation
B) Specification of entire application at very high level
C) Only data entry
D) Only screen design

Correct = B

482. Screen generator is an interactive facility for:
A) Generating code
B) "Painting" screens and performing automatic input validation
C) Testing
D) Compiling

Correct = B

483. Reusable code modules are stored in:
A) Database
B) Program libraries
C) Compiler
D) Linker

Correct = B

484. Application packages are designed for:
A) General purpose only
B) Specific purpose of use (e.g., Tally, Crystal Reports)
C) Operating systems only
D) Hardware only

Correct = B

485. Personal computers with software are used for:
A) Only gaming
B) Application prototype development
C) Only word processing
D) Only calculations

Correct = B

486. According to Sommerville, software engineering is important because:
A) Hardware is expensive
B) More and more society relies on advanced software systems
C) Programming is easy
D) Documentation is optional

Correct = B

487. In Sommerville's view, the four fundamental activities common to all software processes are:
A) Planning, coding, testing, deployment
B) Specification, development, validation, evolution
C) Analysis, design, implementation, maintenance
D) Requirements, design, coding, testing

Correct = B

488. Heterogeneity in software engineering refers to:
A) Same type of systems
B) Systems operating as distributed systems across different types of computers
C) Single platform systems
D) Homogeneous environment

Correct = B

489. According to Sommerville, security and trust in software engineering means:
A) Software should be fast
B) Malicious users cannot successfully attack software and information security is maintained

C) Software should be cheap
D) Software should have many features

Correct = B

490. Sommerville identifies which software product types?
A) Only generic products
B) Generic products and customized (bespoke) software
C) Only customized
D) Only system software

Correct = B

491. Essential attributes of good software according to Sommerville include:
A) Only functionality
B) Acceptability, dependability and security, efficiency, maintainability
C) Only speed
D) Only cost

Correct = B

492. In Sommerville's view, software engineering ethics includes:
A) Only technical skills
B) Confidentiality, competence, intellectual property rights, computer misuse
C) Only programming skills
D) Only testing skills

Correct = B

493. The ACM/IEEE Code of Ethics has how many principles?
A) 5
B) 8
C) 10
D) 12

Correct = B

494. According to Sommerville, agile methods are characterized by:
A) Heavy documentation
B) Interleaved specification, design, and implementation; incremental development; extensive tool support
C) Sequential phases
D) No customer involvement

Correct = B

495. The agile manifesto values all EXCEPT:
A) Individuals and interactions over processes and tools
B) Working software over comprehensive documentation
C) Contract negotiation over customer collaboration
D) Responding to change over following a plan

Correct = C

496. In Scrum, the product backlog is:
A) A list of completed tasks
B) A list of "to do" items for the Scrum team
C) A testing report
D) A design document

Correct = B

497. In Scrum, a sprint is typically:
A) 1 day
B) 2 to 4 weeks
C) 6 months
D) 1 year

Correct = B

498. According to Sommerville, requirements engineering process includes:
A) Only elicitation
B) Elicitation, analysis, specification, validation, management
C) Only specification
D) Only validation

Correct = B

499. In Sommerville's view, system modeling using UML helps to:
A) Write code faster
B) Represent system from different perspectives: external, interaction, structural, behavioral
C) Test the system
D) Deploy the system

Correct = B

500. According to Sommerville, configuration management activities include:
A) Only version control
B) Version control, system building, change management, release management
C) Only change management
D) Only release management

Correct = B